RAKSHIT JAIN

EMBEDDED FIRMWARE ENGINEER

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SUMMARY

Embedded Firmware engineer with experience in Hardware Validation in pre-silicon and post-silicon validation of MCU-based silicon, focusing on UART, I2C, I2S, XSPI, and USB 2.0 peripherals. Hands-on with embedded C firmware test development, register-level debug, Python automation, Linux-based driver/firmware validation, Saleae logic analyzer, oscilloscope measurements, and protocol debugging. Seeking embedded firmware development roles.

SKILLS

Embedded/Firmware: C, Embedded C, C++, ARM Cortex-M, ARM TrustZone, FreeRTOS, STM32, LPC2129, Make, CMake, GDB, GNU LD

Protocols/Interfaces: UART, I2C, SPI, I2S, XSPI, USB 2.0, JTAG, Wi-Fi, TCP/IP

Security/Cryptography: DES encryption/decryption fundamentals; currently learning AES, ECC, and post-quantum cryptography (PQC)

Validation/Debug: Pre-silicon validation, post-silicon validation, FPGA, Veloce, OpenOCD, JTAG IR/DR scans, Saleae Logic Analyzer, oscilloscope, register-level debug, protocol debugging

Scripting/Tools: Python, Bash, Tcl, Linux, Git, STM32CubeMX, ARM GNU Toolchain

Secondary: Flutter, React, Node.js, Firebase, Django

EXPERIENCE

Synaptics — Hardware Validation Engineer **March 2024 – Present**

- Performed pre-silicon validation on FPGA/Veloce and post-silicon validation for MCU peripherals and silicon features.
- Ported FreeRTOS and Trustzone implementation for security access validation on the post-silicon.
- Developed embedded C firmware test cases for UART, I2C, I2S, XSPI, and USB 2.0 peripheral validation.
- Patched OpenOCD to issue undocumented custom JTAG instruction-register (IR) and data-register (DR) scans, enabling debug of signals exposed for USB ATE validation.
- Debugged peripheral bring-up issues using register logs, Saleae logic analyzer captures, oscilloscope measurements, and protocol-level analysis.
- Built Python-based automation routines to validate hardware peripherals and reduce repeated manual test execution.
- Validated Wi-Fi firmware/Linux driver functionality on silicon, including Berkeley Packet Filter and WPA Enterprise test bring-up.

PROJECTS

UART-based Python-like Interpreter — LPC2129, Embedded C **github.com/ra-jain/p-interpreter**

- Built a UART-driven command interpreter on the NXP LPC2129 with serial input handling, command parsing, and calculator-style operations.

C-based File-backed Database — Linux, C **github.com/ra-jain/c-db**

- Built a linked-list database in C with file backup and restore support.

C Preprocessor Tool — Linux, C **github.com/ra-jain/preprocessor**

- Built a C preprocessor utility to remove comments and process header includes.

INTERNSHIP

Sasya Systems — Software Developer Intern **Jan 2023 – June 2023**

- Built Flutter-based applications for car stock auditing and real-time video conferencing.
- Worked with Firebase, Node.js backend services, and team-based deployment workflows.

EDUCATION

- Bachelor of Technology in Computer Science and Engineering, MIT Ujjain — 2023; GPA: 7.82/10